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Effect of intervertebral disc degeneration on mechanical and electric signals at the interface between disc and vertebra

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ABSTRACT

Intervertebral disc (IVD) degeneration is significantly correlated with the changes in structure and material properties of adjacent vertebral bone, possibly through mechanical and electrical interactions. However, the mechanisms underlying the alteration of the mechanical and electrical environment at the disc-vertebra interface related with disc degeneration have not been well studied. The objective of this study was to numerically investigate the long-term distributions of mechanical and electrical signals on the disc-vertebra interface with disc degeneration. A three-dimensional finite element model of a human lumbar IVD was used to study the mechanical and electric signals at the interface between disc and vertebral body. The disc degeneration was simulated by reducing the nutrition levels on the nucleus pulposus (NP)-vertebra interface and on the annulus fibrosus (AF) periphery to 30% and 60% of its normal values, respectively. In the simulation, the total external mechanical load applied to the disc-vertebra segment was assumed unchanged during disc degeneration. The simulation results showed that the compressive stress of solid matrix changed by up to ~37 kPa on the NP-vertebra interface, while it increased by up to ~32 kPa on the AF-vertebra interface. The shear stress increased by up to ~37 kPa with disc degeneration. The absolute value of the electric potential on the disc-vertebra interface of the disc slightly decreased with the disc degeneration (~0.5 mV). The knowledge of these spatial and temporal variations of the mechanical stresses and electric potential on the disc-vertebra interface is important for understanding the vertebrae adaptation and remodeling during disc degeneration.

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1. Introduction

Intervertebral disc (IVD) degeneration is found to be significantly correlated with the changes in structure and material properties of adjacent vertebrae (Harada et al., 1998; Homminga et al., 2012; Simpson et al., 2001; Sornay-Rendu et al., 2006). For example, disc degeneration affects the strength distribution of the adjacent vertebral body, as the stiffness of the trabecular bone adjacent to the nucleus pulposus (NP) was lower as discs degenerate (Keller et al., 1989). In addition, with disc degeneration, the bone mineral density (BMD) decreases in the trabecular centrum and increases in the peripheral areas of the bone (Harada et al., 1998; Homminga et al., 2012; Simpson et al., 2001).

According to the Wolff's Law (Wolff, 1986), with disc degeneration, the vertebral body adjacent to the discs adapts to the altered mechanical environment. One study (Malinin and Brown, 2007) showed that the decompression of baboon lumbar IVDs with injection of enzymes or with surgery created lesions in the adjacent vertebrae within 24 weeks, as a result of bone adaption to its altered mechanical environment. Generally, disc degeneration is a slow process (Sivan et al., 2006); it is virtually impossible to measure the changes in stress in the IVD and in its adjacent vertebrae in vivo over time. Numerical modeling, on the other hand, may be a plausible alternative to study such a slow change in the mechanical signals at the interface between IVD and vertebral body.

Several researchers have numerically studied the variation of the mechanical loading within the vertebral body with the healthy and degenerated discs (Dai, 1998; Homminga et al., 2012; Kurowski and Kubo, 1986). Results of these studies show that in a healthy IVD, the highest stress was located in the center of the vertebral body, whereas in degenerated IVDs, the higher stresses appeared in the outer rims of the vertebral body. Currently, there exists few numerical model capable of describing the temporal changes in the mechanical signals at the disc-vertebra interface associated with the disc degeneration.

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On the other hand, electrical stimulation has been reported to have positive effects on bone tissue growth and repair. Electrical signals have been reported to accelerate bone healing and promote spinal fusion (Khalifeh et al., 2018). It has been found that the electric potential in the disc changes with the alteration of the fixed charge density of the disc (Yao and Gu, 2007). Since the fixed charge density decreases with the disc degeneration (Antoniou et al., 1996; Iatridis et al., 2007; Lyons et al., 1981; Urban and Maroudas, 1979), it is important to know how the electric potential at the disc-vertebra interface varies with the disc degeneration.

The objective of this study was to numerically investigate the changes in mechanical and electrical signals on the disc-vertebra interface with disc degeneration. This study may provide additional information on the interaction between disc degeneration and vertebra adaption.

2. Methods

A three-dimensional finite element model of the human lumbar IVD developed in our previous studies (Gu et al., 2014; Zhu et al., 2014; Zhu et al., 2012) was used to investigate the mechanical and electrical signals at the interface between the disc and the vertebral body (Fig. 1). The model consisted of the IVD and the vertebra: the disc was modeled as an inhomogeneous, isotropic porous mixture consisting of a charged solid phase (collagen-proteoglycan network with cells), a fluid phase (interstitial water), and a solute phase with multiple species (sodium ion, chloride ion, glucose, oxygen, and lactate), whereas the vertebral body was modeled as a single-phase solid. Each phase (or component) was assumed to be intrinsically incompressible and the mixture be electrically neutral (Lai et al., 1991). The mechano-electrochemical signals and cell activities in the disc were coupled (Gu et al., 2014; Lai et al., 1991; Zhu et al., 2012).

The geometry of the disc was obtained from a human adult lumbar IVD (41 y.o., healthy, male, L2-L3) (Jackson et al., 2011). The disc was modeled as an inhomogeneous material with two distinct regions: nucleus pulposus (NP) and annulus fibrosus (AF), and in each region the solid phase was modeled as isotropic and linearly elastic. The constitutive relations for (electro)chemical potentials of fluid and solutes were similar to those in the literature (Lai et al., 1991; Sun et al., 1999).

The initial conditions and mechanical properties were listed in Table 1. More details can be found in (Gu et al., 2014; Zhu et al., 2012). The cartilaginous endplate (CEP) was not considered separately in this study. The concentrations of nutrients at the disc-vertebra interface were adjusted to simulate the effect of endplate on the transport of nutrients to the disc (Table 2) (Gu et al., 2014). The vertebral body was assumed to be a homogeneous, linearly elastic solid with the material properties of λ = 86.5 MPa and μ = 57.7 MPa (Goldstein, 1987). Assuming all signals are symmetrical about the mid-axial plane and the mid-sagittal plane, only a quarter of the disc-vertebra segment was simulated (Fig. 1).

In this study, the disc degeneration was simulated by reducing the nutritional levels at disc boundaries (Gu et al., 2014). The disc

Table 1
List of initial conditions and mechanical properties.

Properties		NP	AF
Initial conditions[a]	Cell density (cells/mm^3)	4000	9000
	Fixed charge density (mol/m^3)	360	360–180
	Oxygen concentration (kPa)	5.1	5.1
	Glucose concentration (mol/m^3)	4	4
	Lactate concentration (mol/m^3)	0	0
	Na^+ concentration (mol/m^3)	150	150
	Cl^- concentration (mol/m^3)	150	150
	Displacement (m)	$u = v = w = 0$	$u = v = w = 0$
	Water content	0.85	0.85–0.70
Mechanical properties	Elasticity constants[b]	λ =400 kPa μ = 100 kPa	λ =400–800 kPa μ =100–200 kPa

[a] From (Zhu et al., 2014; Zhu et al., 2012), the dashed line '-' denotes linear variation from the inner AF to the outer AF.
[b] The elasticity constants are estimated from the aggregate and the shear moduli for human and bovine IVDs reported in the literature (Iatridis et al., 1999; Iatridis et al., 1998; Périé et al., 2006).

Table 2
List of chemical boundary conditions used in this study.

Properties	On AF periphery (Normal/Degenerate)	On NP-vertebra interface (Normal/Degenerate)
Oxygen concentration (kPa)	5.80[a]/3.48	4.08[c]/1.22
Glucose concentration (mol/m^3)	5.00[a]/3.00	3.20[c]/0.96
Lactate concentration (mol/m^3)	1.1[a,b]	1.7[b]
Na^+ concentration (mol/m^3)	150	150
Cl^- concentration (mol/m^3)	150	150

[a] From (Soukane et al., 2007).
[b] Estimated based on (Holm et al., 1981).
[c] Due to the lack of the CEP in our model, all the values reported in (Soukane et al., 2007) were multiplied by 0.8 as an approximation for the boundary conditions on the NP-vertebra interface.

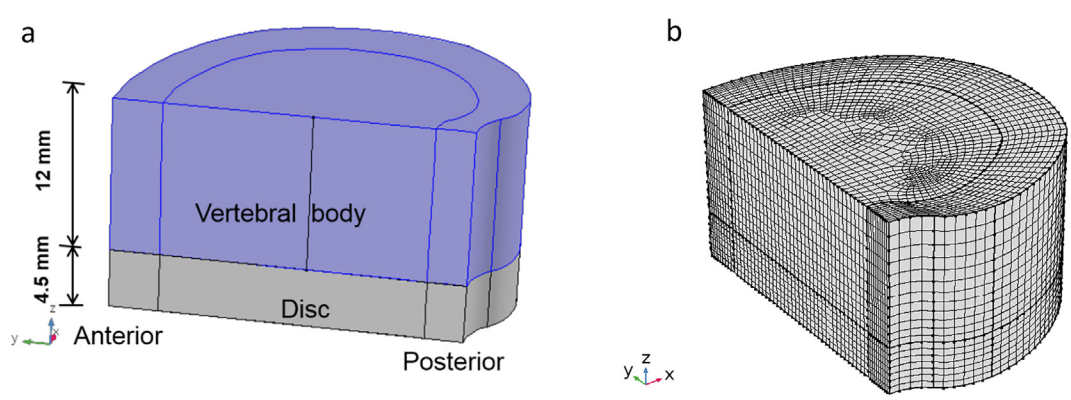

Fig. 1. (a) The schematic of a quarter of a disc-vertebra segment and (b) the mesh of 29,374 hexahedral elements. The total number for degrees of freedom is 840,831.

was first allowed to reach equilibrium with normal nutrition boundary conditions (Table 2) for 40 days. From Day 41 to Day 50, the glucose and oxygen levels on the NP surface (adjacent to the vertebra) were reduced linearly from 100% to 30% of the normal values; and those on the AF periphery were reduced linearly from 100% to 60% of the normal values (Table 2). The boundary conditions then were remained constant for the rest of time.

In this study, the disc at a mature, healthy state (Fig. 1), i.e., before degeneration, was chosen as the reference state. All the results for stresses presented in this study were the difference in stress between the current configuration and the reference configuration (i.e., the change in stress due to disc degeneration). Since the total external mechanical load applied to the disc-vertebra segment was assumed to remain unchanged during disc degeneration, a load-free boundary condition was used on top of the vertebra surface, as well as on the AF peripheral surface. A symmetric boundary condition was used for the stresses, fluid and solutes on the mid-sagittal and mid-axial planes. A zero-flux boundary condition was used for fluid and solutes on the AF-vertebra interface (Gu et al., 2014; Zhu et al., 2012). The electric potential and fluid pressure in the tissue or medium surrounding the disc were assumed to be zero in our simulation.

The simulation was conducted with COMSOL software (Version 5.3, COMSOL Inc., Burlington, MA) based on the finite element method (Sun et al., 1999).

3. Results

The compressive stress of solid matrix decreased on the NP-vertebra interface of the disc, while it increased on the AF-vertebra interface as the disc degenerates (Figs. 2a, 3a, and 4a). The largest increase in the compressive stress was located near outer regions on the AF-vertebra interface, while the largest decrease in the compressive stress was at the center of the NP-vertebra region.

The change in shear stress on the disc-vertebra interface of the disc also varied with disc degeneration, see Figs. 2–4. For the shear stress component τ_{zx}, the largest increase was predicted in the mid-lateral AF region (Fig. 2b, the direction was denoted with black arrows), and noticeably, there was a slight decrease in τ_{zx} in the NP region (Fig. 3b). For the shear stress component τ_{zy}, the largest increase was predicted in the anterior and posterior regions of the disc (shown as the dark red and dark blue region in Fig. 2c, negative value means opposite direction to the y-axis, see also Fig. 4c).

The absolute value of the electric potential on the disc-vertebra interface of the disc decreased as the disc degenerates (Figs. 2d, 3d, and 4d).

4. Discussion

In this study, the mechanical and electrical signals on the disc-vertebra interface of the disc during disc degeneration was investigated numerically. This may be the first report on such a study. As mentioned before, the total mechanical load on the disc-vertebra segment was assumed to be constant over the period of disc degeneration. Our results confirm the notion that with disc degeneration, the mechanical loading is increasingly supported by the AF region (Figs. 3 and 4). This shift in mechanical stress distribution from the NP to AF region is mainly due to the change in fixed charge density distribution in the degenerated disc, which is caused by the non-homogeneous decreases in proteoglycan content therein (Antoniou et al., 1996; Iatridis et al., 2007; Lyons et al.,

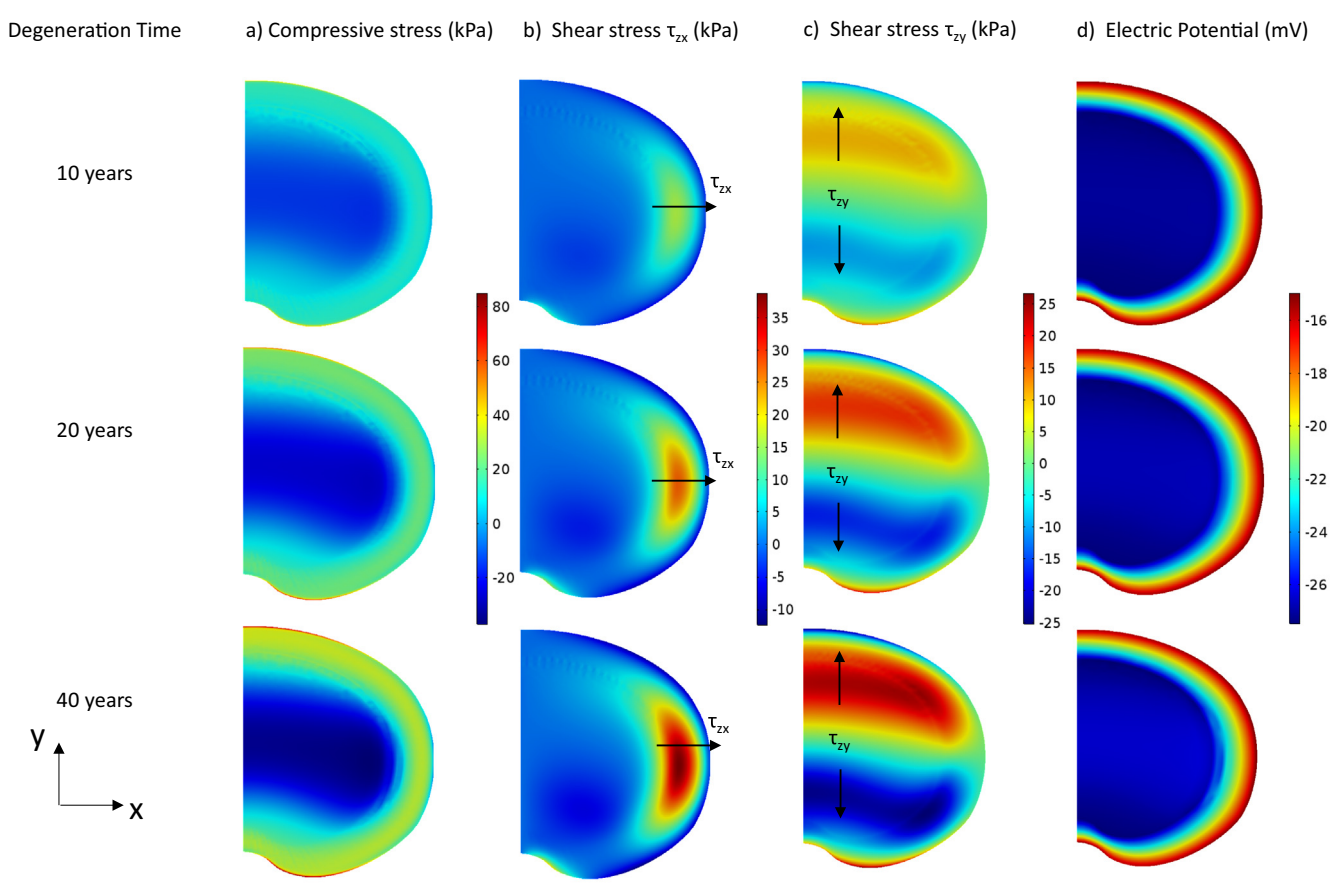

Fig. 2. Changes in the distributions of (a) compressive stress, (b) and (c) shear stresses, and (d) electric potential on the disc-vertebra interface with disc degeneration.

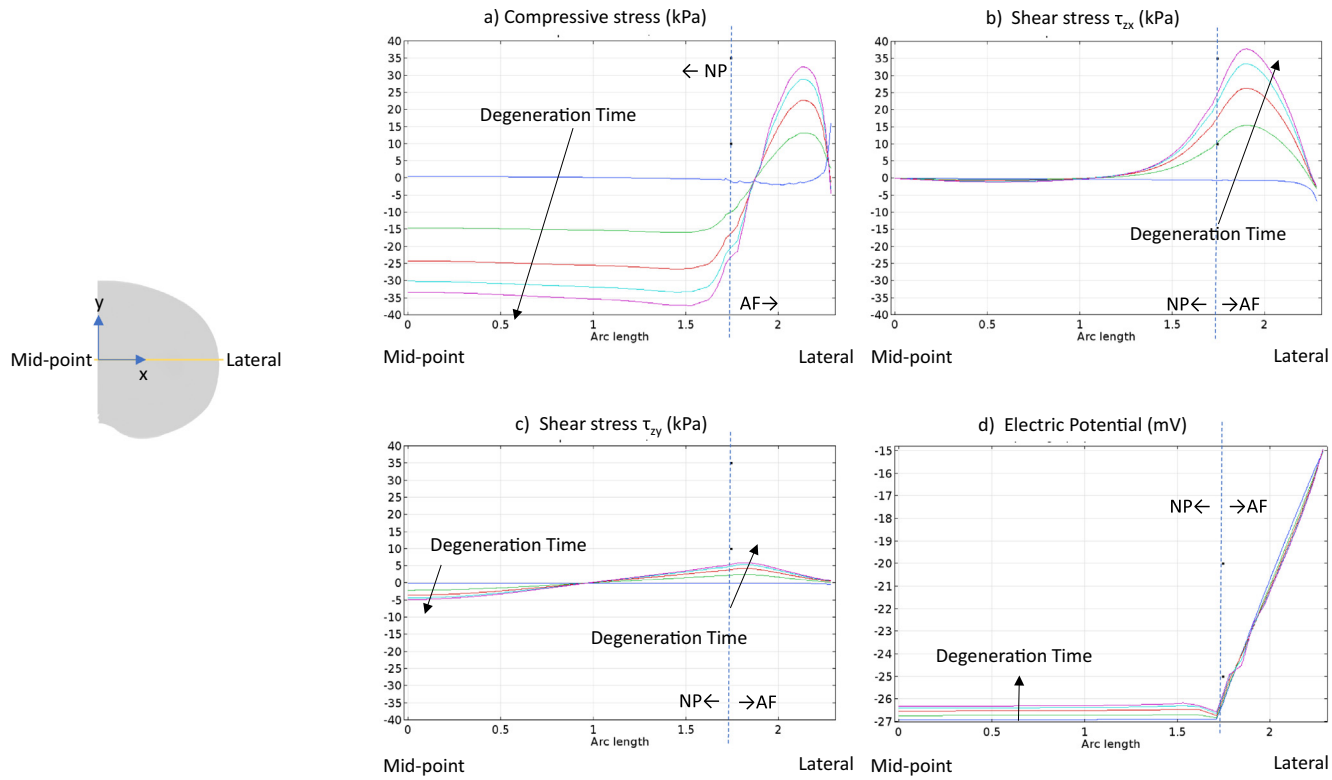

Fig. 3. Changes in the distributions of mechanical and electrical signals on the disc-vertebra interface of the disc (along the coronal direction at y = 0) with disc degeneration over time: (a) compressive stress, (b) and (c) shear stresses, and (d) electric potential. The first line represents the time at Day 40 (when the degeneration starts). Each time increment represents 10 years. The last line represents the time at 40 years.

1981). This result is consistent with the studies on mechanical stresses distribution in the disc (Adams et al., 2000; Adams et al., 1996), as well as on the BMD distribution in the vertebrae (Homminga et al., 2012).

The shear stress on this interface increases with the disc degeneration and points toward the disc periphery. This is because as disc degenerates, the disc shrinks as it loses water due to the reduction of fixed charge density. The adjacent vertebra, however, prevents disc from shrinking in the region close to the vertebral body, thus generating shear stress in the direction toward disc periphery. These findings help understand how vertebrae remodel during disc degeneration.

The electric potential is also related to the fixed charge density in the disc (Gu et al., 1999). As the disc degenerates, the fixed charge density decreases as shown by experiments (Antoniou et al., 1996; Iatridis et al., 2007; Lyons et al., 1981; Urban and Maroudas, 1979) as well as simulations (Yao and Gu, 2007), reducing the electric potential. The electrical signals have long been reported to have positive effects on bone healing, repair and spinal fusion (Khalifeh et al., 2018). Therefore, our simulation provides additional insights to the understanding of vertebral bone fracture and wound healing.

This study does have its limitations. The mechanical properties used in this study are assumed to be independent of time. In other words, no changes in the mechanical properties of the disc and adjacent vertebra with disc degeneration were considered, which may affect the prediction of mechanical stress distribution on the disc-vertebra interface. Moreover, the disc has been assumed to be isotropic, and the solid phase is assumed to be linearly elastic. The model with these simplifications is not able to catch the detailed mechanical information within the collagen-proteoglycan network, such as the anisotropic and fiber-content-dependent stress-strain (i.e., "regional effect") phenomena described in the literature (Derrouiche et al., 2019; Kandil et al., 2019).

Another major limitation is that the cartilaginous endplate (CEP) was not explicitly modeled in our simulations. The CEP is a thin layer of tissue [~0.6 mm in thickness (Roberts et al., 1989)] located in the superior and inferior of the disc (between NP and vertebra). In the current study, the CEP was not explicitly modelled due to the limitation of computational resources. Because the mechanical properties of CEP are not the same as those of NP (Wu et al., 2013), the inclusion of the CEP is expected to affect local stress distributions, especially at its early stage of disc degeneration.

The third limitation is related to the numerical errors in the simulation. The shear stress component, τ_{zy}, at the anterior point in Fig. 4c should be zero because a load-free boundary condition was imposed on the peripheral surface of the disc. However, the predicted value of τ_{zy} at the anterior point is not exactly zero (see Fig. 4c). This is mainly due to the discontinuity in the mechanical properties between the bone and the disc as well as the mesh size. Nonetheless, the region associated with this error is relatively small (see Fig. 4c). Similar phenomena can be seen at lateral point of the disc in Fig. 3b.

For the purpose of reducing computational cost, the disc geometry was assumed to be symmetric in this study (Fig. 1). This simplification may not be applied to most IVDs. It should be pointed out that the degree of disc degeneration depends on the nutrition supply at disc boundaries. With varied nutrition boundary conditions, the simulated results would be different (Zhu et al., 2017; Zhu et al., 2016). Cautions should be taken when interpreting these time-dependent signals reported in this study.

In conclusion, the temporal and spatial distributions of mechanical stress and electric potential at disc-vertebra interface were studied with a 3D finite element model. Both mechanical and

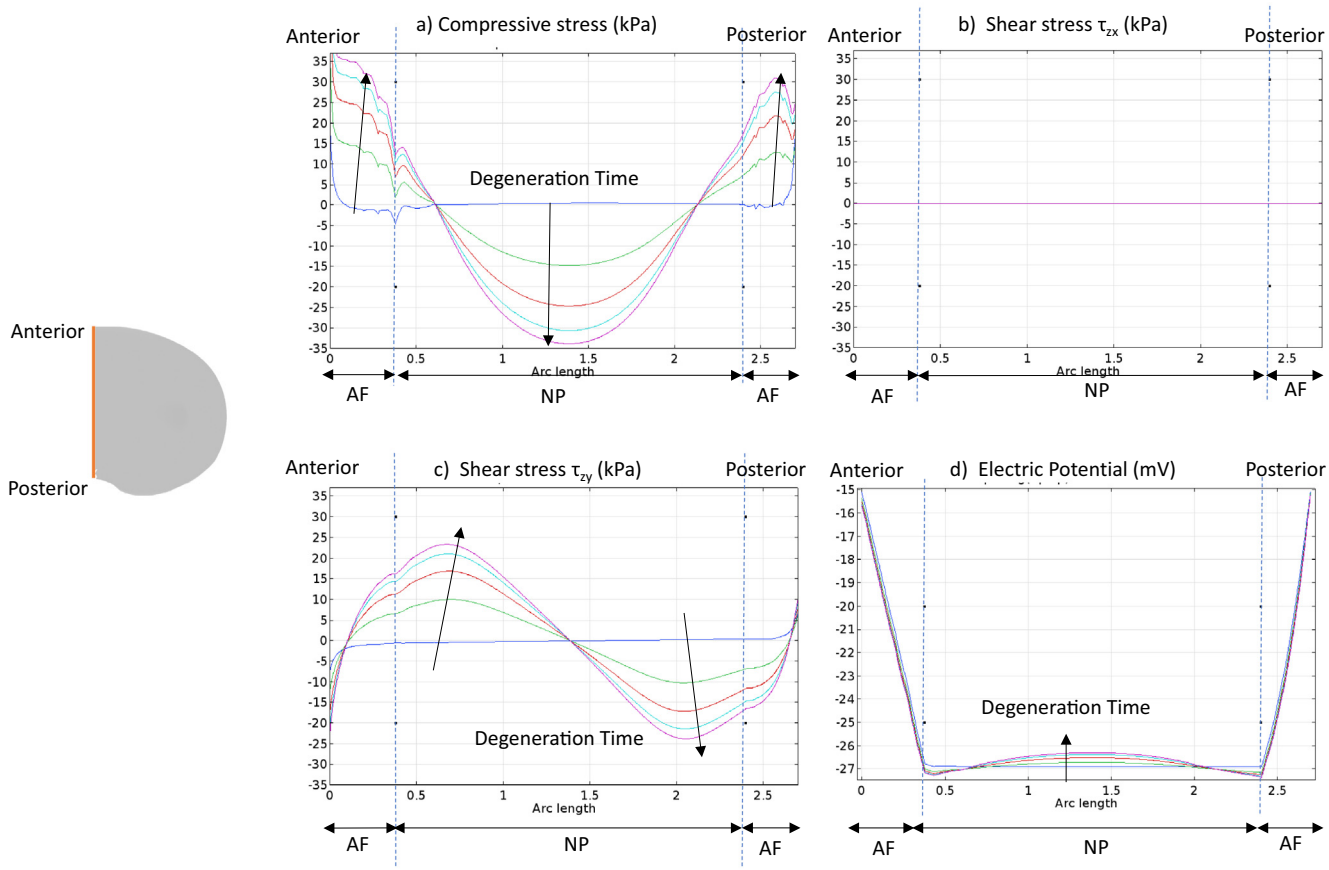

Fig. 4. Changes in the distributions of mechanical and electrical signals on the disc-vertebra interface of the disc (along the sagittal direction at x = 0) with disc degeneration over time: (a) compressive stress, (b) and (c) shear stresses, and (d) electric potential. The first line represents the time at Day 40 (when the degeneration starts). Each time increment represents 10 years. The last line represents the time at 40 years.

electrical signals vary with disc degeneration. The simulation results are consistent with those reported in the literature. The knowledge of these spatial and temporal variations of the mechanical stresses and electric potential at the disc-vertebra interface is important for understanding the vertebrae adaptation and remodeling during disc degeneration.

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Declaration of Competing Interest

The authors declare no conflict of interest.

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